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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1	**Date:** September 16, 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science

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& AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building application	Software Engineering and Backend Developer
Systems & Cloud Infrastructure	Maintaining Hardware provisioning, Network virtualization	System administrator and DevOps engineer
Data Science & AI Engineering	Extracting insights from structured/ data and training predictive AI models	Data scientist and machine learning engineer
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing and managing incident response	Information security analyst and penetration tester
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures	Business system analyst and IT manager

My Chosen Pathway (Part B)

I am most interested in: Software Engineering

Why I chose it:

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I chose it because it sounds good for me to say the word software Engineering. I want also to create an application and to design. Additionally, the job for software Engineering is my dream.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Systems administrators, DevOps Associate	Linux fundamentals, Network basics (IP, DNS), Scripting (Bash, Python), Trouble Shooting & Support, CI/CD Knowledge
Mid-Level	Systems Administrator, Cloud DevOps Engineer	Configuration Management (Ansible, Chef), Infrastructure as Code (Iac, Terraform), Cloud Platforms (AWS, Azure, GCP), Monitoring & Logging (Prometheus, ELK), Network Configuration
Senior-Level	Senior DevOps Engineer, Cloud Architect, Site Reliability Engineer	Advance Cloud Architecture (Scalability, Resilience), Container Orchestration (Kubernetes, Docker), Site Reliability Engineering (SRE) Principles, Incident Response & Disaster Recovery, CI/CD Pipeline Design
Expert / Leadership	DevOps Manager, Director of Engineering, CTO	Organizational Strategy & Vision, DevOps Culture & Process Improvement, Team Leadership & Mentorship, Budgeting & Resource Planning, Business Alignment

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Python	Y	Take advance python course so that I can master it.
System analyst	N	I will enroll in system analyst and study how to improve my skills.
Network security	N	Practice in TryHackme and I will take the cyber security course.
Linux/Unix Admin	N	Learn basic commands through free Linux course.

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Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Database Management	N	I will take database course and I will practice it.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Software Engineer, Backend (All levels)
Company	SIEMENS
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	LinkedIn
Required Skills Listed (3–5)	Strong understanding of C++ programming, including design patterns and algorithms, excellent problem solving, ability to work effectively with multi-functional teams in collaborative environment

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I choose this pathway because as I said that I want to build an application and to be a software Engineer is my dream.

2. Looking at your roadmap, which career stage excites you the most, and why?

The career stage excites me the most is the Expert/leadership stage, because I want to be expert by doing it and I want to share my experience to others if I become an expert of it.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

For me python is i already have or I already experience to do it but not actually an expert sometimes I make some mistakes or errors. The hardest for me to develop is Network security, because I did not already experience and encounter it before.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

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By means of this it confirms my plans because I already wanted to become a software Engineer. Seing the roadmap maid me realize the specific skills I need to focus on during my time here at SSU.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date

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